Ram Calculus and Strict Inverse-First-Power Integral Means

M151: Corrected Tilted-Variance Premises,
Positive-Operator Certificates, and Quantifier-Safe Extensions

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Abstract

The manuscript *A strict inverse-first-power bound for univalent functions* claims a uniform exponent below 1/4, and hence contradicts Kraetzer's proposed quadratic spectrum at $p = -1$. We examine its compact operator, weighted affine law, critical-point pairing, transport, and equality-exclusion dependencies without claiming complete validation of the source proof. We derive exact angular and path-space Ram log-partition identities, retaining the nonzero drift in path coordinates. A substantive correction to an earlier Ram sufficient condition is proved: for the bounded univalent corner map $f(z) = z - z^2/2$, the variance under the -1 tilt is $T^2/12 + O(T)$, although its integral-means exponent is zero. Thus a proposed pointwise tilted-variance bound cannot hold for the full class on an interval containing -1; the averaged identity remains valid. We prove finite-block and positive-supersolution certificates, deterministic residual bounds, and an analytic-tail/coefficient-cover test for the actual univalent-map operator. An exact Koebe-map obstruction rules out unweighted block certificates at every depth 1 through 13. Conditional on a uniform first-power exponent $b < 1/4$, we obtain explicit bounds at neighboring negative exponents and a negative-tail estimate. A conditional transfer theorem derives the analytic bridges for each fixed inverse power $0 < s < 2$, with the global geometric pairing principle left as an explicit source dependency. Its full-family version also has a conditional endpoint exponent consequence by spectral continuity. All independent implications have proofs; no numerical universal gap, sharp first-power lower extremal, full source certification, or unconditional endpoint resolution is asserted.

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1 Problem, Provenance, and Claim Boundaries

1.1 The class and the three growth quantities

Let $\mathcal{S}$ be the normalized univalent maps on $\mathbb{D}$, with $f(0) = 0$ and $f'(0) = 1$, and let $\mathcal{S}_b$ consist of those with bounded image. The image bound may depend on the map. For $p \in \mathbb{R}$ and $0 < r < 1$ put

$$I_f(p,r) = \frac{1}{2\pi}\int_0^{2\pi} |f'(re^{i\theta})|^p \,\mathrm{d}\theta, \quad T(r) = \log\frac{1}{1-r}, \quad \beta_f(p) = \limsup_{r\uparrow 1}\frac{\log I_f(p,r)}{T(r)}. \tag{1}$$

The bounded universal spectrum is

$$B_b(p) = \sup_{f\in\mathcal{S}_b} \beta_f(p). \tag{2}$$

There are no assumed boundary values, boundary smoothness, or common image diameter. Normalizing a *fixed* map does not change its exponent.

Distinguish (2) from the uniform envelope

$$U_{\mathcal{A}}(p) = \limsup_{r\uparrow 1}\frac{\log \sup_{f\in\mathcal{A}} I_f(p,r)}{T(r)}$$

for a declared class $\mathcal{A}$, and from the operator rate $\gamma(s) = \log_2 r(L_s)$ defined below. Supremum and radial limsup cannot generally be interchanged. The source proves a comparison with an operator rate, not an identity between all three quantities.

For clarity the main source conclusion will be used only through the following explicitly declared premise:

$$\begin{gathered} 0 < b < \tfrac{1}{4}, \quad C \geq 1, \qquad I_f(-1,r) \leq C(1-r)^{-b} \\ \text{for every } f \in \mathcal{S} \text{ and } 1/2 \leq r < 1. \end{gathered} \tag{H_b}$$

A result marked conditional on (H_b) is not an independent proof of this premise. It also applies to any smaller class for which that uniform premise has actually been established.

1.2 Pinned source and depth of reading

M151 means manifest id 151 in the local 722-manuscript project. The source is OpenAI's September 24, 2026 manuscript [1], recorded at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Its `thm:main` asserts (H_b) for some $b < 1/4$, without giving a numerical gap. All seven main section files and the bibliography were read in the preceding audit. The critical algebra, operator-rate logic, measurability requirements, ordered transport limits, and harmonic equality barrier were examined. No decisive gap was identified in that bounded inspection. This is *not* full source proof validation, a formal verification, or a statement of community acceptance.

The private input is TR-item-077-astra [2], dated September 6, 2026, read in full. Its exact log-partition and covariance identities, conditional closure criteria, diagonal counterexample, tilted-variance identity, and analytic-tail requirements are the relevant material. Reports referenced inside it are not silently imported as independently reread theorems. M149's completed report was read for formatting and evidence standards; no M149 theorem is a mathematical dependency of this report. M150's inverse-square claim is not used.

1.3 What is and is not improved

The results proved here have three different statuses.

1. Independent elementary results on the stated domains: angular and path partition identities, the critical-tilt counterexample, spectral certificate implications, analytic-tail cover bounds, and power stability.

2. Conditional consequences of (H_b): an averaged-variance deficit, a negative-tail bound, and a two-sided interval about $p = -1$ strictly below the conjectural parabola.

3. A conditional parameter-transfer theorem: the affine law, escape estimates and transport implications are derived, while the global topological and measurable pairing principle remains an explicit source dependency. A full-family hypothesis also gives an endpoint exponent limit.

The new correction in the first category invalidates a proposed *premise*, not the earlier theorem saying that the premise would be sufficient. No result here improves the source's unknown numerical value of b, identifies $B_b(-1)$, or supplies a sharp lower extremal. Classical positive-operator and probability arguments are not claimed as historically new because they are expressed in Ram coordinates.

2 Source Dependency Audit

2.1 The compact model

Use the classical Koebe estimates [3, Chapter 2]

$$\frac{1-|w|}{(1+|w|)^3} \le |f'(w)| \le \frac{1+|w|}{(1-|w|)^3}, \qquad B(0,1/4) \subset f(\mathbb{D}). \tag{3}$$

Let

$$\mathcal{K} = \{F : \mathbb{H} \to \mathbb{C} \text{ univalent} : F(i) = 0,\ F'(i) = 1\}, \quad Q_F = 1/F', \quad m(z) = \frac{z-i}{z+i}.$$

Then $F = 2if \circ m$ identifies $\mathcal{K}$ with $\mathcal{S}$. The locally uniform topology on $\mathcal{K}$ is compact metrizable: distortion gives local boundedness, Montel gives subsequences, and Hurwitz plus the retained normalization preserves nonconstant injectivity. Differentiation and reciprocal differentiation are continuous on compact source sets. This is not a compactness assertion for bounded-image maps with a common image bound.

The affine group and its normalized action are

$$\begin{gathered} z \cdot w = x + yw, \quad z = x + iy, \qquad T_z F(w) = \frac{F(z \cdot w) - F(z)}{yF'(z)}, \\ T_w(T_z F) = T_{z \cdot w} F, \qquad Q_{T_z F}(w) = Q_F(z \cdot w)/Q_F(z). \end{gathered} \tag{4}$$

For $z_\pm = \pm 1/2 + i/2$, set

$$A_z \Phi(F) = |Q_F(z)| \Phi(T_z F), \qquad L = \tfrac{1}{2}(A_{z_-} + A_{z_+}).$$

With $v = 2^{-n}$ and $u_j = -1 + (2j+1)v$, the source obtains

$$L^n = v \sum_{j=0}^{2^n - 1} A_{u_j + iv}, \qquad \|L^n\| = \max_{F \in \mathcal{K}} (L^n 1)(F), \qquad \beta = \log_2 \lim_n \|L^n\|^{1/n}. \tag{5}$$

We keep its notation β for this single operator rate only in this section. It is $\gamma(1)$ later, not $\beta_f(-1)$.

2.2 The dependencies that produce strictness

Source locator	Role in the proof	Audit boundary
`01-compact-model`, `lem:compact`, `prop:disk`	Uniform strip control and disk-to-dyadic comparison.	Local compactness alone is not boundary compactness or a supremum–limsup exchange.
`02-weighted-law`, `prop:law`	Resolvent eigenmeasure and horizontal/scale averages at the same operator exponent.	Requires the exact weighted identity for Borel tests, not only sampled moments.
`03-escape`, `lem:area-tail`, `prop:compact-levels`	Almost-everywhere compact positive superlevels.	Uses global injectivity, target area, and boundary escape; cannot be replaced by finite matrix algebra.
`04-strict-pairing`, `lem:good`, `lem:count`, `prop:pair`	Regular tie-free fibers, index counts and measurable equivariant rank pairing.	All exceptional sets and the exclusion of the pole from finite maxima matter.
`05-strict-transport`, `lem:base-good`, `prop:strict-transport`	Weighted goodness at a fixed target and strict comparison of paired masses.	The three limits are ordered; no uniform positive pairing ratio is asserted.
`06-equality-barrier`, `lem:no-zero-J`	Equality forces an impossible harmonic lower bound.	Excludes equality for the operator exponent, not separately for each map.

Under the contrary assumption $\beta \geq 1/4$, the source constructs a probability P on $\mathcal{K}$ with

$$\mathbb{E}_P[|Q_F(x+iy)|\Phi(T_{x+iy}F)] = y^{-\beta}\mathbb{E}_P\Phi(F) \tag{6}$$

for nonnegative Borel tests. Its resolvent construction gives a measure eigenvector without requiring a positive continuous eigenfunction or a compact operator. The dyadic boundary terms vanish as $O(2^{-n\beta})$; horizontal averaging and a logarithmic scale-period average then give (6).

Writing $q = iyQ'/Q = a + ic$, $k = (\beta+2)/3$, and $G = F - iyF'/k$, set $J = \det DG/|F'|^2$. Differentiation gives

$$\begin{gathered} \mathbb{E}a(i) = -\beta, \qquad \mathbb{E}|q(i)|^2 = \beta(\beta+1), \\ k^2 J = k(k-1) + (2k-1)a + |q|^2, \qquad k^2\mathbb{E}J(i) = \frac{(4\beta-1)(\beta+2)}{9}. \end{gathered} \tag{7}$$

At critical points of $V_\xi = y^k/|F-\xi|$, $G(z) = \xi$ and $V_\xi(z) = ky^{k-1}|Q(z)|$. The Hessian of $\log V_\xi$ has negative trace and determinant $-k^4J/y^4$: positive J means saddle, negative J means finite maximum.

The source's weighted area tail is $\int_{|m(z)|\geq 1/2} y^\beta |F(z)|^{-3}\,\mathrm{d}A < \infty$ almost surely. Together with a horizontal preimage-length bound and a summable $O(h^{2k-1})$ target cover, it prevents escape. Regular fibers and distinct critical values allow a strict rank pairing from each saddle to a higher finite maximum. These are substantive geometric dependencies; the Ram representations below do not prove them again.

2.3 The fixed target and the equality mechanism

Because $G_F(i) = -i/k$ for every normalized map, area-generic goodness of ξ says nothing directly about this target. The source instead proves

$$\mathbb{E}[J_+(i)\mathbf{1}_{\mathrm{bad}(F,G_F(i))}] = 0$$

by first applying the area formula over a source box and then (6). The expected spatial density is $\mathrm{d}x\,\mathrm{d}y/y$. This Jacobian-weighted statement is exactly what is needed.

For the good positive-J domain D, the pairing ratio $c_F(z) = (V(R_F(z))/V(z))^3$ exceeds one. With relative displacement truncated by N, box widths S and heights Y, the source proves

$$2S \log Y \, \mathbb{E}[J_+ c_F \mathbf{1}_{D_N}] \le 2(S + NY)(\log Y + 2\log N)\mathbb{E}J_-.$$

First $S \to \infty$, then $Y \to \infty$ at fixed N, and finally $N \to \infty$ give $\mathbb{E}[J_+ c_F \mathbf{1}_D] \le \mathbb{E}J_-$. Thus $\mathbb{E}J \le 0$. If equality holds, the nonnegative random variable $J_+(c_F - 1)\mathbf{1}_D$ has expectation zero, so weighted goodness implies $J_+ = J_- = 0$ almost surely. No uniform lower bound on $c_F - 1$ is needed.

Lemma 2.1 (Harmonic obstruction used by the source)**.** *For $k = 3/4$, no $F \in \mathcal{K}$ has $J_F \equiv 0$.*

Proof. Equation (7) gives $|q + 1/4|^2 = 1/4$, hence $|F''/F'| \ge 1/(4y)$. The analytic function F''/F' is nowhere zero, so $u = \log|F''/F'|$ is harmonic and $u \ge -\log 4 - \log y$. On $[-1,1] \times [\epsilon, 1]$, $0 < \epsilon < 1/2$, the harmonic function

$$b_\epsilon(x,y) = -\log 4 + \log(1/\epsilon)\cos(\pi x/2)\frac{\sinh(\pi(1-y)/2)}{\sinh(\pi/2)}$$

is below u on the boundary. On the sides and top its added term is zero; on the bottom its factor lies between zero and one. The minimum principle gives $u(i/2) \ge b_\epsilon(0, 1/2)$, whose right side diverges as $\epsilon \downarrow 0$. This is impossible. □

Assuming the preceding source dependencies, (7) and transport force $\beta = 1/4$ and $J(i) = 0$ almost surely. Apply (6) to $|J(i)|$, then use its strictly positive weight and equivariance: $J(z) = 0$ almost surely for each fixed z. A countable dense set and continuity give $J \equiv 0$, contradicting Lemma 2.1. Therefore the single operator rate is $< 1/4$. Choosing $\beta < b < 1/4$ yields $\|L^n\| \le C_b 2^{bn}$, including the finitely many small n, and the disk comparison gives (H_b). This explains the uniform-gap logic without asserting complete verification of all earlier dependencies.

3 Angular Ram Calculus with the Quantifiers Retained

3.1 Exact tilted identities

For a fixed f, r, let $X(\theta) = \log|f'(re^{i\theta})|$ and $\Lambda(p) = \log \mathbb{E}_\theta e^{pX}$. Since f' is nonzero in $\mathbb{D}$, an analytic logarithm exists. Normalization gives $\mathbb{E}_\theta X = 0$.

Theorem 3.1 (Angular log-partition identity)**.** *For each $f \in \mathcal{S}$, $r < 1$, and real p,*

$$\Lambda(p) = p^2 \int_0^1 (1-t)\,\mathrm{Var}_{\nu_{tp}} X \,\mathrm{d}t, \qquad \mathrm{d}\nu_s = e^{sX - \Lambda(s)}\frac{\mathrm{d}\theta}{2\pi}. \tag{8}$$

Moreover β_f and B_b are convex and vanish at zero.

Proof. At fixed radius X is continuous and bounded. Differentiation under the integral gives $\Lambda'(s) = \mathbb{E}_{\nu_s} X$ and $\Lambda''(s) = \mathrm{Var}_{\nu_s} X$. Taylor's integral formula for $t \mapsto \Lambda(tp)$, with $\Lambda(0) = \Lambda'(0) = 0$, gives (8), including negative p. Holder gives $I_f(ap + (1-a)q, r) \le I_f(p,r)^a I_f(q,r)^{1-a}$ for $0 \le a \le 1$. Take logarithms, divide by T, then take limsups to prove convexity of β_f. Taking the supremum preserves convexity. At zero the integral is one. □

For dyadic $r_j = 1 - 2^{-j}$, write $X_j = X_{f,r_j}$ and $d_j = X_j - X_{j-1}$. Since $X_0 = 0$,

$$\operatorname{Var} X_J = \sum_{j=1}^{J} \operatorname{Var} d_j + 2 \sum_{1 \le i < j \le J} \operatorname{Cov}(d_i, d_j). \tag{9}$$

This follows by expanding the centered square. The same identity holds under any one specified tilted measure, but one cannot mix covariance entries from different tilts and still call the sum a variance.

Lemma 3.2 (Dyadic limits and fixed-map lower certificates)**.** *For fixed f, p, the limsup in* (1) *equals its dyadic limsup. If for each $\epsilon > 0$ there is a fixed $f_\epsilon \in \mathcal{S}_b$ and a sequence $r_j \uparrow 1$ with $\limsup_j \log I_{f_\epsilon}(p, r_j)/T(r_j) \ge b - \epsilon$, then $B_b(p) \ge b$.*

Proof. The function $|f'|^p = \exp(p \operatorname{Re} \log f')$ is subharmonic, so its circular means increase with r. Jensen gives $I_f(p, r) \ge 1$. For $r_j \le r \le r_{j+1}$ the logarithm of the integral is between the corresponding endpoint values. Since $T(r_{j+1})/T(r_j) \to 1$, sandwiching proves the first assertion. For the second, the full radial limsup of a fixed f_ϵ dominates its subsequence limsup. Take the supremum over maps and then $\epsilon \downarrow 0$. □

3.2 An exact diagonal failure

For $0 < a < 1$, $f_a(z) = z/(1 - az)$ is normalized, bounded and univalent. For fixed a, its derivative and reciprocal are bounded up to the circle, so $\beta_{f_a}(p) = 0$ for all real p. Angular orthogonality gives

$$I_{f_a}(2, r) = \sum_{k \ge 0} (k+1)^2 a^{2k} r^{2k} = \frac{1 + a^2 r^2}{(1 - a^2 r^2)^3}.$$

On the diagonal $a = r \uparrow 1$, this has exponent three. Thus even within this family the supremum of fixed-map exponents is zero while the diagonal envelope has positive growth. The known $B_b(2) = 1$ [7, Theorem 1.8] makes the contrast with the whole bounded class sharper. An optimizer changing both the map and the scale is not an extremal construction for (2).

4 A Genuine Correction to the Earlier Ram Premise

4.1 What the old condition said

TR-item-077, equations (15)–(16), correctly proved (8) and proposed the sufficient condition

$$\sup_{|s| \le p_0} \frac{\operatorname{Var}_{\nu_s} X_{f,r}}{T(r)} \le \frac{1}{2} + \epsilon_f(r), \qquad \epsilon_f(r) \to 0, \tag{10}$$

for every fixed bounded univalent f. It would imply $B_b(p) \le p^2/4$ on $[-p_0, p_0]$. That implication is valid. However, for $p_0 \ge 1$ the premise itself is false, not merely unproved.

Theorem 4.1 (Critical corner tilt)**.** *Let $f(z) = z - z^2/2$ and $T = \log(1/(1-r))$. Then $f \in \mathcal{S}_b$, $\beta_f(-1) = 0$, and under the probability density proportional to $|f'(re^{i\theta})|^{-1}$,*

$$\mathbb{E}_{\nu_{-1}} X = \tfrac{1}{2} \log(1 - r^2) = -\frac{T}{2} + O(1), \qquad \operatorname{Var}_{\nu_{-1}} X = \frac{T^2}{12} + O(T). \tag{11}$$

In particular (10) *fails for this single map whenever $p_0 \ge 1$.*

Proof. Write $f = [1-(1-z)^2]/2$. The numbers $1-z$ lie in the open right half-plane. If their squares agree, the numbers are equal or negatives; the latter cannot both lie in that half-plane. Thus f is injective. It is a bounded normalized polynomial, and $f' = 1-z$.

Put $\delta = 1-r = e^{-T}$ and

$$D_r(\theta) = |1-re^{i\theta}| = \sqrt{\delta^2 + 4r\sin^2(\theta/2)}, \qquad M_j(r) = \int_0^\pi \frac{\log^j D_r(\theta)}{D_r(\theta)}\,\mathrm{d}\theta, \quad j = 0,1,2.$$

We claim

$$M_j(r) = \frac{(-1)^j}{j+1}T^{j+1} + O(1+T^j). \tag{12}$$

Fix a small $\theta_0 \in (0,1/2)$ and take $r \geq 1/2$. On $[0,\delta]$, D_r is comparable to δ and $\log D_r = -T + O(1)$, giving $O(1+T^j)$. On $[\theta_0,\pi]$ the integral is bounded independently of r. For $\delta \leq \theta \leq \theta_0$ put $e(\theta) = \delta + \theta^2 + (\delta/\theta)^2$. The identity for D_r and the Taylor estimate for sine give uniform positive upper and lower bounds for D_r/θ, as well as

$$\left|\frac{D_r}{\theta} - 1\right| + |\log(D_r/\theta)| \leq Ce(\theta).$$

Consequently the difference between the integrand in M_j and $\theta^{-1}(\log\theta)^j$ is at most $Ce(\theta)\theta^{-1}(1+|\log\theta|^j)$. Its integral is $O(1+T^j)$: the three terms are bounded respectively by $O(\delta(1+T^{j+1}))$, $O(1)$, and $O(1+T^j)$. Integrating $\theta^{-1}(\log\theta)^j$ from δ to θ_0 proves (12).

Symmetry cancels the factor two in normalized tilted moments. Thus $I_f(-1,r) = M_0/\pi = (T+O(1))/\pi$, whose logarithm divided by T tends to zero. Also

$$\mathbb{E}_{\nu_{-1}}X = M_1/M_0 = -T/2 + O(1), \qquad \mathbb{E}_{\nu_{-1}}X^2 = M_2/M_0 = T^2/3 + O(T).$$

Subtracting the squared mean gives (11). Dividing its variance by T gives divergence, proving the failure of (10).

For the exact mean, put $a = 1-r^2$ and use the circle involution $w = (r-z)/(1-rz)$, $|z| = 1$. Direct differentiation gives

$$1 - rw = \frac{a}{1-rz}, \qquad |\,\mathrm{d}w| = \frac{a}{|1-rz|^2}|\,\mathrm{d}z|, \qquad \frac{|\,\mathrm{d}w|}{|1-rw|} = \frac{|\,\mathrm{d}z|}{|1-rz|}.$$

Thus the tilted probability is invariant and $X(w) = \log a - X(z)$. Its mean is exactly $\frac{1}{2}\log a$; indeed its centered distribution is reflection symmetric for every $0 < r < 1$. All the moments exist at each fixed radius, so this substitution needs no limiting interchange. □

The corrected research target is the *averaged* integral in (8), not a uniform pointwise variance bound over the entire tilt interval. A large variance at the critical tilt is consistent with a small integral-means exponent; an endpoint peak cannot be replaced by a uniform bound inferred from its weighted integral.

The earlier cumulant criterion also needs its stated analytic convergence and remainder hypotheses. When the series is valid at $p = -1$, write

$$\frac{\Lambda(-1)}{T} = \frac{\kappa_2}{2T} + \frac{\mathcal{R}_3(-1)}{T}.$$

A strict upper bound controls this sum. Only with the additional premise $\mathcal{R}_3(-1) = o(T)$ may one conclude $\limsup \kappa_2/T \leq 2b$. The present corner counterexample refutes (10); it does not refute every possible cumulant upper criterion. Finally, if (H_b) holds, the old matching lower-certificate premise for $B_b(-1) = 1/4$ is impossible. That equality goal must not be silently retained as a compatible objective.

5 The Actual Transfer Operator as a Ram Partition Function

5.1 Power operators and their paths

For every real s define on $C(\mathcal{K})$

$$A_{z,s}\Phi(F) = |Q_F(z)|^s \Phi(T_z F), \qquad L_s = \tfrac{1}{2}(A_{z_-,s} + A_{z_+,s}). \tag{13}$$

The weights are strictly positive and continuous on compact $\mathcal{K}$. For independent fair reference signs $\sigma_1, \dots, \sigma_n$, set

$$F_0 = F, \quad F_j = T_{z_{\sigma_j}} F_{j-1}, \quad d_j = \log |Q_{F_{j-1}}(z_{\sigma_j})|, \quad S_n = \sum_{j=1}^{n} d_j.$$

Independence of the signs does not imply independence of these state-dependent increments.

Theorem 5.1 (Path partition and the missing drift)**.** *For every F, n, s,*

$$\Psi_n(F,s) := \log \mathbb{E}_\sigma e^{sS_n} = \log(L_s^n 1)(F), \tag{14}$$

$$\Psi_n(F,s) = s\mathbb{E}_\sigma S_n + s^2 \int_0^1 (1-t)\, \mathrm{Var}_{\pi_{ts}} S_n \, \mathrm{d}t, \tag{15}$$

where π_u is the path probability tilted by e^{uS_n}. Moreover, with v, u_j as in (5)*,*

$$L_s^n = v \sum_j A_{u_j + iv, s}, \qquad \gamma(s) := \lim_n \frac{\max_F \Psi_n(F,s)}{n \log 2} = \log_2 r(L_s). \tag{16}$$

Proof. The cocycle (4) telescopes the product of path weights to $|Q_F(z_{\sigma_1} \cdots z_{\sigma_n})|^s$. Averaging the 2^n paths proves (14). Their affine products have height 2^{-n} and abscissae $\sum_{\ell=1}^n \pm 2^{-\ell}$, each occurring once; this proves the first identity in (16). At fixed n the distribution is finite, so differentiating its partition function twice gives tilted variance. Taylor's formula gives (15); the first derivative at zero is $\mathbb{E}_\sigma S_n$, which is not generally zero. For a positive operator on $C(\mathcal{K})$, $|L_s^n \Phi| \le L_s^n 1$ when $\|\Phi\|_\infty \le 1$, with equality in the norm at $\Phi = 1$. Thus $\|L_s^n\| = \max_F (L_s^n 1)(F)$. Submultiplicativity gives the root limit and its infimum characterization; the spectral-radius formula gives the last equality. □

There is a simple exact drift check. For $f(z) = z$, the corresponding $F = 2im$ has $|Q_F(z_\pm)| = |z_\pm + i|^2/4 = 5/8$. Hence at $n = 1$, $S_1 = \log(5/8)$ on both paths: $\Psi_1(F,s) = s\log(5/8)$ and the tilted variance is zero. An angular-centering argument applied to this path partition would incorrectly predict zero. The angular and path identities are different representations with different reference measures. Nor is the weighted affine law (6) an unweighted stationary law for the fair-sign process.

5.2 Disk comparison for the actual operator

Proposition 5.2 (Uniform comparison at each fixed power)**.** *For each $s \ge 0$ there is $C_s < \infty$ independent of f, n, r such that*

$$I_f(-s, r) \le C_s \, \|L_s^n\|, \qquad 2^{-n} \le 1 - r < 2^{1-n}, \quad r \ge 1/2. \tag{17}$$

Thus $B_b(-s) \le \gamma(s)$.

Proof. For the disk arc $3\pi/4 \le \theta \le 5\pi/4$, its inverse Cayley coordinates $z = x + iy = m^{-1}(re^{i\theta})$ satisfy

$$x = \frac{-2r\sin\theta}{d}, \quad y = \frac{1-r^2}{d}, \quad d = 1 - 2r\cos\theta + r^2.$$

They obey $|x| \le 1$, $1/2 \le y/v \le 4$ and $\mathrm{d}x/\,\mathrm{d}\theta \ge 1/16$. In the cell $|x - u_j| \le v$, the rebased point $(x-u_j)/v + iy/v$ stays in one fixed compact rectangle in $\mathbb{H}$. Compact-family bounds and the cocycle give $|Q_F(x+iy)| \le C|Q_F(u_j+iv)|$. The exact identity $|f'(re^{i\theta})|^{-1} = 4|Q_F(x+iy)|/|z+i|^2$ has denominator bounded below. Raise to s, integrate over cells of length $2v$, and use $\mathrm{d}\theta \le 16\,\mathrm{d}x$ to obtain $C_s v \sum_j |Q_F(u_j+iv)|^s$. Four rotations of this arc cover the circle, and the rotated disk maps remain in $\mathcal{S}$. Equation (16) proves (17). Finally divide logarithms by T and use $n/T \to 1/\log 2$, first for each fixed map and then for the supremum. □

6 Quantitative Spectral Certificates with Full Proofs

6.1 Finite blocks, supersolutions and residuals

Theorem 6.1 (Two equivalent routes with slack)**.** *For the exact $L = L_1$, the following are equivalent:*

1. *$r(L) < 2^{1/4}$;*
2. *for some integer $N \ge 1$, $\left\|L^N\right\| \le A < 2^{N/4}$;*
3. *for some $h \in C(\mathcal{K})$ and $0 < m_h \le h \le M_h < \infty$, $Lh \le \lambda h$ on all $\mathcal{K}$ with $0 < \lambda < 2^{1/4}$.*

A certificate in (2) gives

$$b_N = \frac{\log_2 A}{N}, \qquad \|L^n\| \le C_N 2^{b_N n}, \quad C_N = \max_{0\le r<N} \|L^r\|\, 2^{-b_N r}. \tag{18}$$

A certificate in (3) gives $\|L^n\| \le (M_h/m_h)\lambda^n$.

Proof. The root limit proves (1)⇒(2). Write $n = qN + r$ and use submultiplicativity to obtain (18), so (2)⇒(1). For (2)⇒(3), put $\lambda = A^{1/N}$ and $h = \sum_{j=0}^{N-1} \lambda^{-j} L^j 1$. It is continuous and at least one, hence bounded above on $\mathcal{K}$, and

$$Lh = \lambda(h-1) + \lambda^{1-N} L^N 1 \le \lambda h.$$

For (3), $1 \le h/m_h$ and positivity imply $L^n 1 \le \lambda^n h/m_h$, giving the claimed norm bound and (1). Equivalently, it gives a finite-block certificate for any $\lambda' \in (\lambda, 2^{1/4})$ once $(M_h/m_h)\lambda^N \le (\lambda')^N$. No assertion of identical-endpoint equivalence is required. □

The affine map $F(z) = z - i$ shows $\|L^n\| \ge 1$; hence valid block data have $A \ge 1$. Combining (18) with Proposition 5.2 gives a uniform exponent $b_N < 1/4$. If $b_N = 0$, one may choose any larger $b < 1/4$ for the notation (H_b). A numerical certificate has not been supplied here. Existence follows from the source only if its operator gap is accepted.

Proposition 6.2 (Deterministic residual budget)**.** *Suppose $h \in C(\mathcal{K})$, $h \ge m_h > 0$, and a computable approximation satisfies, pointwise on all $\mathcal{K}$,*

$$\widetilde{L}h \le \lambda_0 h + e_{\mathrm{cert}}, \qquad \left\|(L - \widetilde{L})h\right\|_\infty \le e_{\mathrm{op}},$$

with nonnegative errors and $\lambda_0 \ge 0$. Then

$$Lh \le \left(\lambda_0 + \frac{e_{\mathrm{cert}} + e_{\mathrm{op}}}{m_h}\right) h. \tag{19}$$

If the parenthesis is $< 2^{1/4}$, it certifies a strict gap.

Proof. Add the two errors and use $1 \le h/m_h$. Continuity and compactness bound h above, so Theorem 6.1 applies. □

A small residual at selected maps or a small eigenvalue of a finite matrix is not the asserted global residual. One must certify how the finite representation dominates the original operator, including its tail.

6.2 An exact obstruction to short unweighted blocks

Proposition 6.3 (Koebe transient obstruction)**.** *For every integer $1 \le n \le 13$, the exact operator satisfies $\|L_1^n\| > 2^{n/4}$. Consequently any certificate of type (2) in Theorem 6.1 must have $N \ge 14$. This restriction is on unweighted norm blocks, not on weighted supersolutions with a nonconstant h.*

Proof. The Koebe map $f(w) = w/(1-w)^2$ belongs to $\mathcal{S}$. Its half-plane normalization and reciprocal derivative are

$$F(z) = 2if(m(z)) = -\tfrac{i}{2}(z^2+1), \qquad Q_F(z) = i/z.$$

For $v = 2^{-n}$, the dyadic formula therefore gives exactly

$$K_n := (L_1^n 1)(F) = 2\sum_{j=1}^{2^{n-1}} \frac{1}{\sqrt{(2j-1)^2+1}} \le \|L_1^n\|\,. \tag{20}$$

No bounded-image hypothesis is needed: the operator's state space is the whole normalized class $\mathcal{K}$.

Here is a fully integer interval test, including its finite data. Put $D = 10^{14}$ and define

$$a_n = 2\sum_{j=1}^{2^{n-1}} \operatorname{isqrt}\left(\left\lfloor \frac{D^2}{(2j-1)^2+1} \right\rfloor\right), \qquad u_n = a_n + 2^n.$$

For a positive integer d, the integer in this summand is exactly $\lfloor D/\sqrt{d} \rfloor$: both expressions are the largest integer t with $t^2 d \le D^2$. Hence

$$a_n/D \le K_n < u_n/D.$$

The following numerators were independently recomputed by integer arithmetic in preparing this report. A lower test means the exact integer inequality $a_n^4 > 2^n D^4$; the upper test means $u_n^4 < 2^n D^4$. Neither comparison uses floating-point square roots or thresholds.

n	Exact lower numerator a_n	Verified test
1	141421356237308	lower
2	204666909440674	lower
3	272174407715770	lower
4	341010964020940	lower
5	410204263516028	lower
6	479488505049382	lower
7	548795596281882	lower
8	618108407149948	lower
9	687422648378706	lower
10	756737247225782	lower
11	826051935479002	lower
12	895366646083378	lower
13	964681362274692	lower
14	1033996079860890	upper

For $n \le 13$, $K_n \ge a_n/D > 2^{n/4}$, proving the obstruction. At $n = 14$, $K_{14} < 1033996079877274/10^{14} < 2^{14/4}$. This last fact is only about the displayed Koebe map and is not an upper bound for the operator norm. □

The obstruction is finite-depth transient growth, not an asymptotic contradiction to a strict operator gap. Indeed $2/\sqrt{(2j-1)^2+1} = 2/(2j-1) + O(j^{-3})$; the harmonic-sum comparison in (20) gives $K_n = n \log 2 + O(1)$ and therefore $\log_2 K_n/n \to 0$. The state F is fixed in this computation, whereas the operator norm maximizes over all states at each depth. The result also explains why a spectral gap need not be visible in a short raw block.

6.3 Error in a partition field

Proposition 6.4 (Log-partition stability)**.** *On a fixed probability space let $Y, \widetilde{Y}$ be bounded real fields with $\left\|Y - \widetilde{Y}\right\|_\infty \le \eta$. Then*

$$\left|\log \mathbb{E}e^{sY} - \log \mathbb{E}e^{s\widetilde{Y}}\right| \le |s|\eta.$$

For n path increments each approximated with error at most η_0, the corresponding rate error is at most $|s|\eta_0/\log 2$.

Proof. Pointwise exponential comparison bounds the ratio of the two integrals between $e^{-|s|\eta}$ and $e^{|s|\eta}$. Take logarithms. For paths, sum the n errors and divide the resulting bound by $n \log 2$. □

A fixed per-step error therefore does not vanish with depth. The angular version for an approximation of $\log f'$ has rate error $|p|\eta/T$, but only for the same admissible map and a proved all-angle error.

7 A Concrete Analytic-Tail Test for Finite Blocks

7.1 Coefficient covers, not sampled maxima

The preceding spectral criteria become useful only with a method that covers all admissible maps. The following is one explicit sufficient construction. It is independent of the source's strict transport proof.

Fix N, put $v = 2^{-N}$, $z_j = -1 + (2j+1)v + iv$, and $w_j = m(z_j)$. Define

$$r_* = \max_j |w_j| < 1, \quad c_j = \frac{|z_j + i|^2}{4}, \quad \ell_j = \frac{1 - |w_j|}{(1 + |w_j|)^3}.$$

Choose R with $r_* < R < 1$ and degree $d \ge 0$. For $f \in \mathcal{S}$, write $f'(w) = \sum_{k \ge 0} b_k w^k$, where $b_0 = 1$. By (3) and Cauchy,

$$|b_k| \le M_R R^{-k}, \qquad M_R = \frac{1+R}{(1-R)^3}.$$

Let finitely many complex coefficient boxes $\mathcal{B}_\alpha$ cover every vector $(b_0, \dots, b_d)$ arising from $\mathcal{S}$. For each box, specify centers $\widehat{b}_{\alpha k}$ and radii $\eta_{\alpha k} \ge 0$ such that $|b_k - \widehat{b}_{\alpha k}| \le \eta_{\alpha k}$ throughout that box. The cover need not consist of coefficient vectors of univalent maps; it is an *outer* cover. A necessary condition such as $b_0 = 1$ may be imposed exactly.

Theorem 7.1 (Certified finite-block outer bound)**.** *Set*

$$D_{\alpha j} = \sum_{k=0}^{d} \widehat{b}_{\alpha k} w_j^k,$$

$$E_{\alpha j} = \sum_{k=0}^{d} \eta_{\alpha k} |w_j|^k + M_R \frac{(|w_j|/R)^{d+1}}{1 - |w_j|/R},$$
$$a_{\alpha j} = \max\{\ell_j,\ |D_{\alpha j}| - E_{\alpha j}\},$$
$$A_N = 2^{-N} \max_{\alpha} \sum_{j=0}^{2^N-1} \frac{c_j}{a_{\alpha j}}.$$

Then all denominators are positive and $\left\|L^N\right\| \le A_N$*. In particular, a rigorous enclosure* $A_N < 2^{N/4}$ *proves the strict spectral criterion in Theorem 6.1.*

Proof. For every actual f, choose a box containing its coefficient vector. Cauchy's estimate bounds the unrepresented derivative tail by $M_R \sum_{k>d} (|w_j|/R)^k$, and the finite coefficient errors give $|f'(w_j) - D_{\alpha j}| \le E_{\alpha j}$. Therefore $|f'(w_j)| \ge |D_{\alpha j}| - E_{\alpha j}$. Independently Koebe gives $|f'(w_j)| \ge \ell_j > 0$. Their maximum is thus a valid lower bound for each denominator of

$$|Q_F(z_j)| = \frac{c_j}{|f'(w_j)|}.$$

Sum at the *same* coefficient box for all j, maximize over boxes, and then over actual F. Equation (5) proves the result. □

This construction does not certify that a box center is univalent; it does not need to for an upper bound. For a lower certificate, the direction changes and actual univalence would be indispensable. Discarding a box is safe only when one proves it contains no admissible coefficient vector. A finite Grunsky test used without its logical direction or tail conditions cannot justify such a discard.

The simple Cauchy coefficient bounds already give a bounded set that can be covered by finitely many boxes. They may be much too loose to pass the strict test. There is no claim that refining this particular outer model must terminate successfully, even if the source theorem is true. All modulus evaluations, coefficient radii, arithmetic errors, and the maximum must have outward error control in a numerical implementation. A maximum over finitely many *sample maps* instead gives a lower estimate of $\left\|L^N\right\|$ and has the wrong direction.

7.2 The all-scale cost remains visible

Here $r_* = (1 - v)/\sqrt{1 + v^2}$. To verify it, note that $|m(x + iv)|^2 = (x^2 + (1 - v)^2)/(x^2 + (1 + v)^2)$ increases with x^2; the extreme abscissae are $\pm(1 - v)$. For $R = (1 + r_*)/2$, a sufficient degree for tail tolerance $\eta > 0$ is

$$d + 1 \ \ge\ \frac{\log(M_R/[\eta(1 - r_*/R)])}{\log(R/r_*)} \tag{21}$$

when $r_* > 0$ and the numerator is positive. This follows by solving the geometric-tail inequality, not by fitting observed errors. As N grows, a fixed degree gives no uniform boundary-tail control. For fixed η the displayed crude sufficient scale is of order $2^N(N + \log(1/\eta))$. This is a cost of this estimate, not an information- theoretic lower bound or a measured algorithmic speed.

The finite physical Gram issue in TR-item-077 also remains relevant. If $\log f'(z) = \sum_{k\ge1} a_k z^k$, angular orthogonality gives

$$\mathrm{Var}_\theta \,\mathrm{Re} \log f'(re^{i\theta}) = \frac{1}{2} \sum_{k\ge1} |a_k|^2 r^{2k}.$$

On a truncated representation $a = Ru$, the corresponding form is $u^* R^* M_r R u$, with $M_r = \frac{1}{2}\,\mathrm{diag}(r^{2k})$, not $u^* u$. This follows by substitution. It preserves a physical norm, but does not establish global univalence, suppress an omitted tail, or remove the critical tilted variance of Theorem 4.1.

8 Conditional Consequences of a Uniform First-Power Bound

Theorem 8.1 (Averaged variance and negative tails)**.** *Assume* (H_b)*. Uniformly for $f \in \mathcal{S}$ and $r \geq 1/2$,*

$$\int_0^1 (1-t)\frac{\mathrm{Var}_{\nu_{-t}} X}{T}\,\mathrm{d}t \leq b + \frac{\log C}{T}, \tag{22}$$

$$\Pr_\theta\{X \leq -aT\} \leq Ce^{-(a-b)T}, \qquad a > b. \tag{23}$$

These assertions do not imply a pointwise tilted-variance bound or a full large-deviation principle.

Proof. Take logarithms in (H_b) and apply Theorem 3.1 with $p = -1$. For (23), on the event in question $e^{-X} \geq e^{aT}$. Markov's inequality and $\mathbb{E}e^{-X} = I_f(-1, r) \leq Ce^{bT}$ prove the bound. A bound on an integral does not bound each integrand value, as Theorem 4.1 already demonstrates; Markov gives no matching lower probability or LDP. □

Theorem 8.2 (An explicit neighboring interval)**.** *Under* (H_b)*,*

$$B_b(-s) \leq \begin{cases} sb, & 0 \leq s \leq 1, \\ b + s - 1, & s \geq 1. \end{cases} \tag{24}$$

The corresponding finite-radius bounds are uniform over $\mathcal{S}$. In particular,

$$B_b(-s) < s^2/4 \qquad \text{for } 4b < s < 2 - 2\sqrt{b}. \tag{25}$$

This interval contains $s = 1$ and lies strictly below $s = 2$.

Proof. For $0 \leq s \leq 1$, Holder on normalized angular measure gives $I_f(-s, r) \leq I_f(-1, r)^s \leq C^s(1 - r)^{-sb}$. For $s \geq 1$, Koebe gives $|f'|^{-(s-1)} \leq [8/(1 - r)]^{s-1}$, hence $I_f(-s, r) \leq 8^{s-1}C(1 - r)^{-(b+s-1)}$. Taking the fixed-map limsup followed by its supremum proves (24). When $s \leq 1$, $sb < s^2/4$ precisely for $s > 4b$. When $1 \leq s < 2$, $b + s - 1 < s^2/4$ precisely for $(s - 2)^2 > 4b$, or $s < 2 - 2\sqrt{b}$. Since $0 < b < 1/4$, the stated interval contains one and has upper endpoint below two. □

This is a proved conditional extension, not merely continuity rhetoric. Its width is nonnumerical until b is certified. It invokes no inverse-square theorem. At $s = 2$, (24) only gives $B_b(-2) \leq 1 + b$, not the Brennan value one.

9 Power Stability and a Conditional Transport Extension

9.1 A controlled perturbation of the same operator

Proposition 9.1 (Spectral power stability)**.** *Let $r_0 = 1/\sqrt{5}$ and*

$$q_{\min} = \frac{5}{8}\frac{(1 - r_0)^3}{1 + r_0}, \quad q_{\max} = \frac{5}{8}\frac{(1 + r_0)^3}{1 - r_0}, \quad H = \max\{|\log q_{\min}|, |\log q_{\max}|\}.$$

For all real s, t,

$$|\gamma(s) - \gamma(t)| \leq \frac{H}{\log 2}|s - t|. \tag{26}$$

The function γ is convex.

Proof. At $z_\pm$, $|m(z_\pm)| = r_0$ and $|z_\pm + i|^2/4 = 5/8$. Koebe therefore bounds $|Q_F(z_\pm)|$ between the displayed constants for every F. Every rebased state also lies in $\mathcal{K}$, so $|S_n| \le nH$. Pointwise path comparison gives

$$e^{-nH|s-t|} L_t^n 1 \le L_s^n 1 \le e^{nH|s-t|} L_t^n 1.$$

Take norms, logarithms and root limits to prove (26). Holder on the path probability gives $(L^n_{\alpha s+(1-\alpha)t}1)(F) \le (L_s^n 1(F))^\alpha (L_t^n 1(F))^{1-\alpha}$. Maximize, divide logarithms by $n \log 2$, and pass to the limits to obtain convexity. No differentiability of γ or spectral simplicity is assumed. □

The exact constants, not their rounded evaluations, are the certificate. This result controls nearby power tests on the same operator, whereas Theorem 8.2 is a direct angular consequence with a generally more useful explicit interval.

9.2 Correct exponents for a proposed power-family argument

Fix $s > 0$ and a real $\gamma \ge 0$. Suppose a probability P on $\mathcal{K}$ satisfies

$$\mathbb{E}[|Q_F(z)|^s \Phi(T_z F)] = y^{-\gamma} \mathbb{E}\Phi(F) \tag{27}$$

for every fixed $z = x + iy$ and nonnegative Borel Φ. Set $k = (\gamma + s + 1)/(s + 2)$ and retain $q = iyQ'/Q$ and $J_k = \det D(F - iyF'/k)/|F'|^2$.

Proposition 9.2 (Conditional moment and transport algebra)**.** *Under* (27)*,*

$$\begin{gathered} \mathbb{E}\,\mathrm{Re}\, q(i) = -\gamma/s, \quad \mathbb{E}|q(i)|^2 = \gamma(\gamma+1)/s^2, \\ k^2 \mathbb{E} J_k(i) = \frac{(4\gamma - s^2)(\gamma + s + 1)}{s^2(s+2)^2}. \end{gathered} \tag{28}$$

On nonsingular fibers of $G_k = F - iyF'/k$, the source density $y^{\gamma-1}|Q|^s J_{k,\pm}\,\mathrm{d}A$ becomes target weight

$$y^{\gamma-1}|Q|^{s+2} = (V_\xi/k)^{s+2}, \qquad V_\xi = y^k/|F - \xi|.$$

Proof. The constant test gives $\mathbb{E}|Q(x+iy)|^s = y^{-\gamma}$. Since $\partial_y |Q|^s = s|Q|^s \,\mathrm{Re}\, q/y$ and $\Delta |Q|^s = s^2 |Q|^s |Q'/Q|^2$, differentiating at i proves the first two identities. Compact-family bounds justify the differentiations. Insert them into $k^2 J_k = k(k-1) + (2k-1)\,\mathrm{Re}\, q + |q|^2$. On multiplying by $s^2(s+2)^2$, the numerator is $4\gamma^2 + (-s^2 + 4s + 4)\gamma - s^2(s+1)$, which factors as displayed. Change of variables on each nonsingular chart divides by $|J_k||F'|^2$. This adds two powers of $|Q|$. At a critical point $V_\xi = k y^{k-1}|Q|$, and $(s+2)(k-1) = \gamma - 1$, proving the last identity. □

Completing the square also gives the useful pointwise identities

$$\begin{gathered} k^2 J_k = |q + k - 1/2|^2 - 1/4, \qquad 0 \le J_{k,-} \le \frac{1}{4k^2}, \\ \mathbb{E} J_k(i) = \frac{4\gamma - s^2}{s^2(\gamma + s + 1)}. \end{gathered} \tag{29}$$

The first two hold for every map and point; the last uses the law. In particular the negative transport mass has an explicit bound, without a compactness-dependent unspecified constant.

Theorem 9.3 (Explicit conditional equality exclusion)**.** *Fix $0 < s < 2$. Suppose* (27) *holds and the corresponding geometric transport hypotheses have been established sufficiently to give*

$$\mathbb{E} J_k(i) \le 0, \qquad \mathbb{E} J_k(i) = 0 \implies J_k(i) = 0 \quad P\text{-almost surely.}$$

Then $\gamma < s^2/4$.

Proof. If $\gamma \geq s^2/4$, Proposition 9.2 and the sign condition force equality $\gamma = s^2/4$ and $J_k(i) = 0$ almost surely. Now $k = (s+2)/4 \in (1/2, 1)$. The weighted law applied to $|J_k(i)|$, equivariance and a countable dense set give $J_k \equiv 0$ almost surely. The Jacobian identity rewrites as $|q + (k - 1/2)| = 1/2$, implying $|q| \geq 1 - k = (2-s)/4 > 0$. Thus $|F''/F'| \geq (1-k)/y$. The proof of Lemma 2.1, with $-\log 4$ replaced by $\log(1-k)$, gives a contradiction. □

The preceding theorem states an exact implication, but by itself leaves transport as a large premise. We now derive the parameter-dependent analytic bridges and isolate a narrower geometric dependency. This is a source-template transfer theorem, not an unconditional announcement.

9.3 The exact law at general power

Lemma 9.4 (Parameter-dependent affine law)**.** *Fix $s > 0$ and let $\gamma = \gamma(s)$. If $\gamma > s - 1$, there is a probability P satisfying* (27)*.*

Proof. The compact strip estimate $|Q_F(x+iy)| \leq C/y$ for $|x| \leq 2$, $y \leq 1$, together with the affine test map, gives $1 \leq \|L_s^n\| \leq C_s 2^{sn}$; in particular $0 \leq \gamma \leq s$. Write $\rho = 2^\gamma$. For $\tau > \rho$ put $R_\tau = \sum_{n \geq 0} \tau^{-n} L_s^n$, $g_\tau = R_\tau 1$, $M_\tau = \max g_\tau$. Norm convergence and positivity give

$$L_s g_\tau = \tau(g_\tau - 1) \leq \tau(1 - 1/M_\tau) g_\tau, \qquad \rho \leq \tau(1 - 1/M_\tau).$$

Thus $M_\tau \geq \tau/(\tau - \rho) \to \infty$. At a maximizer F_τ, $\mu_\tau = \delta_{F_\tau} R_\tau / M_\tau$ is a probability and $\mu_\tau L_s = \tau \mu_\tau - (\tau/M_\tau)\delta_{F_\tau}$. Compactness of probabilities on $\mathcal{K}$ gives a weak subsequential limit $\mu L_s = \rho \mu$.

The composition law $A_{z,s} A_{w,s} = A_{z \cdot w, s}$ and the dyadic grid give

$$\mu A_{2+i,s} - \mu = \rho^{-n} v (\mu A_{1+v+iv,s} - \mu A_{-1+v+iv,s}).$$

Its norm is at most $C_s \rho^{-n} v^{1-s} = C_s 2^{-n[\gamma - (s-1)]} \to 0$. Consequently $\mu A_{2+i,s} = \mu$, and $x \mapsto \mu A_{x+ih,s}$ has period two for every fixed $h > 0$.

Set $\bar{\mu} = \frac{1}{2} \int_0^2 \mu A_{x+i,s} \, \mathrm{d}x$ on continuous tests. Periodicity gives horizontal invariance, and changing variables in the two summands of L_s gives $\bar{\mu} A_{i/2,s} = \rho \bar{\mu}$. Hence $h^\gamma \bar{\mu} A_{ih,s}$ is multiplicatively two-periodic. The finite positive functional

$$\nu = \int_1^2 h^\gamma \bar{\mu} A_{ih,s} \frac{\mathrm{d}h}{h}$$

satisfies $\nu A_{iy,s} = y^{-\gamma} \nu$ by logarithmic substitution and periodicity. Horizontal invariance follows from $A_{ih,s} A_{x+i,s} = A_{hx+i,s} A_{ih,s}$. Its mass is finite and strictly positive by compact-family continuity. Normalize ν to obtain P. The identities initially hold on continuous tests. Both sides are finite weighted-pushforward Borel measures on compact $\mathcal{K}$, so equality extends to every nonnegative Borel test, including unbounded ones by monotone convergence. This proves (27). □

9.4 The area tail and both escape regimes

Lemma 9.5 (Escape at power s)**.** *Suppose $s > 0$, $\gamma \geq 0$ and* (27) *holds. Put $a = \gamma + s - 1$ and $k = (\gamma + s + 1)/(s+2) > 1/2$. For P-almost every F,*

$$I_s(F) := \int_{|m(z)| \geq 1/2} y^a |F(z)|^{-(s+2)} \, \mathrm{d}A(z) < \infty.$$

For almost every such F, and then area-almost every ξ, every positive closed superlevel of $V_\xi = y^k/|F - \xi|$ is compact in $\mathbb{H}$.

Proof. For $z \neq i$, let $z^{\#} = (-x+i)/y$. Apply (27) to $|F(z^{\#})|^{-(s+2)}$ and use $(T_z F)(z^{\#}) = -F(z)Q_F(z)/y$. This gives

$$\mathbb{E}[|F'(z)|^2 |F(z)|^{-(s+2)}] = y^{-\gamma-s-2}\mathbb{E}|F(z^{\#})|^{-(s+2)}.$$

On $S_0 = \{|m(z)| \geq 1/2\}$, the quarter theorem gives $|F(z)| \geq c > 0$. Injectivity bounds the integral of the left side by $\int_{|\omega|\geq c} |\omega|^{-(s+2)}\,\mathrm{d}A(\omega) = 2\pi c^{-s}/s < \infty$. The involution $z \mapsto z^{\#}$ preserves S_0, inverts height, and has absolute Jacobian y^{-3}. Changing variables on the right gives precisely $\mathbb{E}I_s(F) < \infty$. Tonelli proves the first assertion.

Fix such an F. On $B(z, y/2)$, heights are comparable to y. For centers with $|x| + y \to \infty$, these disks eventually lie in S_0 and escape every bounded $|x|+y$ region. The submean inequality for $|1/F|^{s+2}$ gives, because $(s+2)k = a+2$,

$$\left(\frac{y^k}{|F(z)|}\right)^{s+2} \leq C_{s,\gamma} \int_{B(z,y/2)} (\operatorname{Im} w)^a |F(w)|^{-(s+2)}\,\mathrm{d}A(w) \longrightarrow 0.$$

This comparison is valid even when $a < 0$. Thus compact target sets have bounded preimage height and compact preimage within every strip $y \geq \epsilon > 0$.

For a compact target set choose smooth nonnegative compactly supported ψ dominating its indicator, and put $H(y) = \int_{\mathbb{R}} \psi(F(x+iy))\,\mathrm{d}x$. At every fixed $u > 0$ the support above $u/2$ is compact. Differentiation and two integrations by parts yield

$$H(u) = \int_{y>u} (y-u)|F'(z)|^2 (\Delta\psi)(F(z))\,\mathrm{d}A(z).$$

If Y_ψ bounds the preimage heights of $\operatorname{supp}\psi$, injectivity bounds its absolute value by $Y_\psi \int_{\mathbb{C}} |\Delta\psi|\,\mathrm{d}A$, independently of u. This proves the horizontal preimage-length bound needed at the boundary.

For omitted targets $|\xi| \leq R$ and a fixed $\delta > 0$, partition a dyadic band $h \leq y \leq 2h$ into horizontal cells of width h. A witness with $V_\xi(z) \geq \delta$ satisfies $y|F'(z)| \leq 4|F(z) - \xi| \leq 4y^k/\delta$ by the local quarter theorem. Compact rebasing on $[-1,1] + i[1/2,1]$ therefore gives $|F(x'+ih) - \xi| \leq C_{k,\delta} h^k$ throughout its cell. Every marked cell maps its whole height-h segment into one fixed compact target disk. The horizontal length bound gives at most C/h marked cells. Disks of radius $C_{k,\delta}h^k$ centered at their midpoint images cover the witnessed targets with total area $O(h^{2k-1})$. Since $2k-1 = (2\gamma+s)/(s+2) > 0$, these areas sum over dyadic scales. The limsup target set is null. Taking integer R and reciprocal-integer δ proves $\sup_x V_\xi(x+iy) \to 0$ as $y \downarrow 0$ for almost every omitted target. No measurability of a witness choice is required: the countable cells determine ordinary target disks.

For ξ in the image, a sufficiently small closed target disk has compact inverse image under the conformal homeomorphism. Below its minimum height, $|F - \xi|$ is uniformly positive, giving the same boundary decay. At heights bounded below the infinity estimate gives $|F| \to \infty$ and $V_\xi \to 0$. Finally $\{V_\xi \geq \delta\} = \{|F - \xi| \leq y^k/\delta\}$ is relatively closed, including its possible pole. The two escape exclusions make it compact. □

9.5 The precise geometric dependency

Definition 9.6 (Planar pairing principle $\mathsf{G}(k)$)**.** For fixed $k > 0$, a good pair (F, ξ) has compact positive superlevels of V_ξ, a regular value ξ of G_k, and no tied critical values. The principle asserts the following deterministic statements on $\mathcal{K}$:

1. The good-pair set is Borel and invariant under affine rebasing. If a fixed F has compact positive superlevels for area-almost every target, almost every target is good.

2. On $D = \{(F, z) : J_k(z) > 0,\ (F, G_k(z)) \text{ good}\}$ there is a Borel equivariant pairing $R_F(z)$ to a finite maximum in the same target fiber. It is injective on each fiber and $V(R_F(z)) > V(z)$.

This is the source's Section 4 geometric input, stated separately instead of assumed to follow from operator algebra. Its displayed calculations are parameter compatible: $\Delta \log V = -k/y^2 < 0$, the Hessian determinant is $-k^4 J_k/y^4$, and a pole is not a preimage of G_k. The difference of critical log values on two distinct inverse branches has nonzero target differential by injectivity of F; this excludes ties outside a null set for any fixed $k > 0$. For regular compact superlevels, the planar count

$$\#\text{components} = \#\text{finite maxima} - \#\text{saddles} + \#\text{poles}$$

has no $k \le 1$ restriction. At most one pole gives the required rank inequality. The source's countable local inverse branches are available at nonzero J_k, and common positive rescaling of critical values preserves ranks. These checks identify no parameter-dependent topological obstruction, even if $k > 1$ before the equality reduction. Nevertheless the full global topology and Borel rank-selection principle is kept as an explicit source dependency here, not declared independently certified by these checks.

Theorem 9.7 (Conditional transfer for every fixed inverse power)**.** *Fix $0 < s < 2$. Suppose $\mathsf{G}(k)$ holds for each*

$$\frac{s+2}{4} \le k \le \frac{2s+1}{s+2}.$$

Then $\gamma(s) < s^2/4$. For every $\gamma(s) < b_s < s^2/4$ there is $C_{s,b_s} < \infty$ such that

$$I_f(-s, r) \le C_{s,b_s}(1-r)^{-b_s} \quad (f \in \mathcal{S},\ 1/2 \le r < 1). \tag{30}$$

No numerical gap or uniform gap as $s \uparrow 2$ is asserted.

Proof. Suppose $\gamma = \gamma(s) \ge s^2/4$. The norm bounds in Lemma 9.4 give $\gamma \le s$ and place $k = (\gamma + s + 1)/(s+2)$ in the stated interval. Since $\gamma - (s-1) \ge (s-2)^2/4 > 0$, that lemma constructs the exact law. Lemma 9.5 gives compact superlevels. By $\mathsf{G}(k)$, good targets are area-generic for almost every map and a Borel equivariant strict pairing is available.

Set $\omega = y^{\gamma-1}|Q|^s$. The area formula in Proposition 9.2 annihilates bad-target positive-Jacobian mass over $|x| \le 1$, $1 \le y \le 2$. Equivariance and (27) give

$$(2\log 2)\mathbb{E}[J_{k,+}(i)\mathbf{1}_{\mathrm{bad}(F,G_k(i))}] = 0.$$

Thus the fixed target $G_k(i) = -i/k$ is dealt with by weighted mass, not by an invalid application of area-genericity to a point.

On D put $c = (V(R)/V)^{s+2} > 1$, and restrict to D_N by $1/N \le y'/y \le N$, $|x' - x| \le Ny$ for the partner $x' + iy'$. These restrictions and c are equivariant. The area formula, exact target weight, and fiberwise injectivity give

$$\int_B \omega J_{k,+} c \mathbf{1}_{D_N}\, \mathrm{d}A \le \int_{B'} \omega J_{k,-}\, \mathrm{d}A,$$
$$B = \{|x| \le S, 1 \le y \le Y\}, \qquad B' = \{|x| \le S + NY, 1/N \le y \le NY\}.$$

Expectation under (27) has density $\mathrm{d}x\,\mathrm{d}y/y$, so

$$2S\log Y\, \mathbb{E}[J_{k,+}(i)c(i)\mathbf{1}_{D_N}] \le 2(S+NY)(\log Y + 2\log N)\mathbb{E}J_{k,-}(i).$$

For this fixed k, $J_k(i)$ is bounded on $\mathcal{K}$, and $\mathbb{E}J_{k,-}(i) \le 1/(4k^2)$ by (29). First let $S \to \infty$, then $Y \to \infty$ for fixed N, then $N \to \infty$ by monotone convergence. Weighted base-goodness and $c > 1$ imply

$\mathbb{E}J_k(i) \le 0$, with equality only when $J_k(i) = 0$ almost surely: indeed $b_+ = \mathbb{E}J_{k,+} \le A = \mathbb{E}[J_{k,+}c\mathbf{1}_D] \le d = \mathbb{E}J_{k,-}$; if $b_+ = d$, then $\mathbb{E}[J_{k,+}(c-1)\mathbf{1}_D] = 0$, so $b_+ = d = 0$. All expectations used in this last comparison are finite because $A \le d$.

The hypotheses of Theorem 9.3 are now established under $\mathsf{G}(k)$, giving the contradiction $\gamma < s^2/4$. For any larger b_s below $s^2/4$, the root limit gives $\|L_s^n\| \le C2^{b_s n}$ for all n. Proposition 5.2 proves (30). □

The conditional theorem proves all of its analytic and probabilistic bridges above while naming the remaining source geometric principle. It is stronger than a formal moment calculation but is not labeled an unconditional universal theorem. A separate local parameter audit (Appendix A) read the full main source and independently checked the law, escape, topology's parameter requirements, transport and equality chain. It found no additional parameter-dependent obstruction, including when $k > 1$ before equality reduction. This is corroborating local review, not formal certification or external validation of $\mathsf{G}(k)$. The factor $2\pi c^{-s}/s$ in the area estimate and the boundary sum can deteriorate as $s \downarrow 0$. As $s \uparrow 2$, the horizontal cancellation margin $(s-2)^2/4$ and the equality-barrier constant $(2-s)/4$ vanish. These are genuine parameter-uniformity limits for the direct proof, not a prohibition on a separate endpoint exponent argument using continuity.

Corollary 9.8 (Conditional endpoint exponent, not an endpoint prefactor)**.** *Suppose the geometric hypotheses of Theorem 9.7 hold for a sequence $s_j \uparrow 2$ in $(0,2)$. Then*

$$\gamma(2) = 1, \qquad B_b(-2) = U_{\mathcal{S}}(-2) = 1.$$

For every $\epsilon > 0$ there is $C_\epsilon < \infty$ with

$$I_f(-2,r) \le C_\epsilon (1-r)^{-(1+\epsilon)} \quad (f \in \mathcal{S},\ 1/2 \le r < 1).$$

These conclusions are conditional on the stated family of geometric dependencies, not consequences of the single first-power premise (H_b).

Proof. Theorem 9.7 gives $\gamma(s_j) < s_j^2/4$. Proposition 9.1 gives continuity, so $\gamma(2) \le 1$. For each $\epsilon > 0$, the root limit bounds $\|L_2^n\| \le C'_\epsilon 2^{(1+\epsilon)n}$ at every integer n; Proposition 5.2 gives the asserted radial estimate. It follows that $B_b(-2) \le U_{\mathcal{S}}(-2) \le 1$. For the bounded univalent corner map of Theorem 4.1, orthogonality of the geometric series gives

$$I_f(-2,r) = \frac{1}{2\pi}\int_0^{2\pi} |1 - re^{i\theta}|^{-2}\,\mathrm{d}\theta = \sum_{n\ge 0} r^{2n} = \frac{1}{1-r^2}.$$

Consequently $B_b(-2) \ge 1$ and, by the disk-to-operator comparison, $\gamma(2) \ge 1$. All three rates are therefore one. □

No uniform control of C_{s,b_s} as $s \uparrow 2$ was used: the limit was taken in spectral rates first, then the root-limit estimate was applied directly to L_2. The result does not give $I_f(-2,r) \le C(1-r)^{-1}$ with a universal constant, boundedness of $2^{-n}L_2^n$, or strictness at $s = 2$. The explicit corner map already rules out a strict endpoint exponent below one. This distinction also explains why failure of the direct equality barrier at $s = 2$ is consistent with the conditional endpoint conclusion.

10 Literature with Declared Reading Depth

The public-primary checks below were performed on October 7, 2026 in the preceding M151 audit and reused here. They are not an exhaustive search for corrections or independent endorsements. Earlier

papers' statements that a conjecture is open are dated context, not disproofs of a later claim. Private report passages were not used as web queries.

Hedenmalm–Shimorin [4], Section 6, Proposition 6.9, Remarks 6.10–6.11 and its implementation table were inspected. The $p = -1$ upper entry 0.403 lies above $1/4$; their full Bergman proof and optimization were not reaudited. Sola's publisher abstract [5] confirms a three-term improvement. The 0.388 value is attributed to Sola by M151 but was not independently checked in the inaccessible full numerical table during this audit. Neither value validates or refutes the later strict bound.

Beliaev's introduction, operator discussion and Table 1 [6] were inspected. His particular snowflake operators supply lower-bound information and distinguish averaged growth from individual-domain spectra. Some numerical entries lack full error estimates. They cannot be reversed into the global upper certificate of Theorem 6.1.

Jin's real-spectrum paper [7], version 12, was checked at its definitions and Theorems 1.1, 1.8, 1.12 and 1.14–1.16. It records $B_{\mathcal{S}}(p) = \max\{B_b(p), 3p-1\}$ and a supremum over bounded quasiconformally extendible maps, together with continuity statements. The equality of bounded and unrestricted per-map universal spectra at negative exponents does not identify either with γ. In the complex-exponent paper [8], version 13, Conjecture 1.4, Theorem 1.15, Theorem 6.20 and Corollaries 6.22–6.23 were inspected. Strict sub-universality for each quasiconformally extendible map is not a uniform common gap; rational-map formulas give $\beta_R(-1) = 0$. Brennan for that rational subclass is not the universal endpoint theorem.

Duplantier–Gayrard–Saksman [9], Theorems 1.1–1.2 and Propositions 1.3–1.4, were checked. Their random-zeta and holomorphic-chaos primitives exhibit the Kraetzer-form spectrum but are not injective. They do not provide admissible lower extremals for B_b. Zheng [10], Theorems 1.3–1.4 and Section 2's reduction, was read at statement/reduction depth: simply connected polynomial basins and small nonconstant, nonleading coefficient perturbations remain subclass hypotheses. Its full proof was not audited. Ivrii [11] was checked at abstract/status depth only for the asymptotic-variance context, not as a proof of an all-map $p = -1$ bound.

The earlier quadratic central law is therefore a conjectural target, not an external theorem to which every representation must conform. The principal unconditional correction in this report, Theorem 4.1, follows directly from an explicit map and does not depend on choosing between the new manuscript and the older conjecture.

11 Advantages, Limitations, and a Testable Program

11.1 What the Ram formulation contributes

The angular partition identifies exactly which negative tilts matter. The path partition uses the source's actual rebasing operator and keeps its drift and cross-scale covariance. Positive supersolutions turn a proposed numerical improvement into a one-sided inequality with an explicit acceptance threshold. The coefficient-cover theorem provides a domain-correct route from analytic reconstruction to that threshold, without declaring finite coordinate algebra to be univalence.

The corner computation is a concrete improvement in the research premises: it prevents an impossible pointwise-variance program at $p = -1$ while preserving the useful averaged criterion. Conditional interval and tail bounds give consequences that can be checked independently of the source's hard measure/topology argument. These benefits are clarity and valid certificate targets, not a measured speedup or an additional universal spectral bound.

11.2 Costs and boundaries

A cover of all admissible coefficients can be very large. Koebe floors and Cauchy tails are rigorous but often loose. Discarding impossible boxes requires genuine necessary conditions with correct error direction. Positive supersolutions may be difficult to find and certify globally. Local compactness does not bound the discontinuous rank pairing's excursions or provide a uniform positive ratio above one. The source's exact strictness is nonquantitative; that is not itself a proof gap, but it leaves numerical engineering work unresolved.

No finite selection of radii proves an all-scale assertion without a block, monotonicity, or uniform-error argument. For an upper spectrum bound, per-map errors tending to zero can suffice when the final order is $\sup_f \limsup_r$. For (H_b), all-map uniform constants are required. For a lower spectrum bound, a fixed map for each tolerance is essential. These are three distinct acceptance conditions.

11.3 Next steps with pass/fail criteria

1. Search for an unweighted block of depth at least 14, or a continuous positive trial function. Proposition 6.3 rules out depths 1–13, but does not establish feasibility at 14. Accept a universal numerical improvement only after a global upper enclosure gives $A < 2^{N/4}$ or $\lambda_0 + (e_{\mathrm{cert}} + e_{\mathrm{op}})/m_h < 2^{1/4}$. A failed enclosure is inconclusive, not a counterexample.

2. Develop sharper admissible coefficient outer bodies and correlated derivative enclosures in Theorem 7.1. Retain every actual map; record the analytic tail and arithmetic budgets separately.

3. Replace the disproved pointwise condition (10) by direct control of the averaged tilt integral. Test borderline corner maps before assuming uniform regularity in the tilt variable.

4. Seek fixed-map lower constructions below $1/4$, with true univalence and infinite-scale bounds. Random noninjective primitives and moving-map optimizers are only diagnostics.

5. Independently certify the geometric principle $\mathsf{G}(k)$ across the full interval in Theorem 9.7, including $k > 1$. Keep the conditional endpoint exponent limit separate from the failed direct endpoint barrier and from an exact endpoint prefactor estimate.

6. Independently recheck the source's Sections 3–5 as one connected measure, topology and selection argument. A rederivation of the polynomial in (7) alone is not that validation.

12 Conclusion

M151 has direct core relevance to TR-item-077, but relevance is not validity. The source's proposed strict gap concerns one uniform operator rate, so its logic cannot be dismissed as a per-map strictness argument. The present report supplies independent, fully proved certificate implications and a substantive correction: the earlier pointwise tilted-variance premise fails for a bounded univalent corner map. Exact integer bounds also eliminate the first thirteen unweighted block depths as possible strict certificates. Exact averaged and path partition representations remain useful when their measures, drift, domains and quantifiers are preserved. A parameter-transfer theorem proves the analytic bridges for fixed $0 < s < 2$ under a named geometric principle. Under its full-family hypotheses, continuity also gives the endpoint exponent one, without claiming an endpoint prefactor bound or promoting the source dependency to an independently validated theorem. A certified numerical gap and a complete validation of the source proof remain outside the results established here.

A Exact Source Locators and Evidence Limits

The local manifest is

C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/manuscripts.json.

The complete main-source root, denoted M/ here, is

C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/openai-math/preprints/A-strict-inverse-first-power-bound-for-univalent-functions-September-24-2026/build/.

Exact file/line locators in the inspected snapshot are:

- M/sections/00-introduction.tex, lines 44–57: thm:main; lines 91–99: no inverse-square theorem input.
- M/sections/01-compact-model.tex, lines 30–68: lem:compact; lines 70–150: eq:action, eq:dyadic, eq:rho, prop:disk.
- M/sections/02-weighted-law.tex, lines 11–94: prop:law; lines 99–125: lem:moments.
- M/sections/03-escape.tex, lines 27–98: lem:area-tail and lem:infinity. Lines 101–148 give prop:compact-levels.
- M/sections/04-strict-pairing.tex, lines 10–50: eq:Jacobian, eq:hessian; lines 54–115: lem:good; lines 119–199: lem:count; lines 211–268: prop:pair.
- M/sections/05-strict-transport.tex, lines 12–52: eq:area-formula, lem:base-good; lines 55–117: prop:strict-transport.
- M/sections/06-equality-barrier.tex, lines 8–43: lem:no-zero-J and its use; lines 44–52: uniform-gap conclusion.

The confidential report path is

C:/Users/mural/OneDrive/Documents/My Documents 2026/res-rt/res-math/TR-item-077-astra.tex.

Its relevant locators are lines 107–146 (log-partition and normalization), 148–171 (covariance), 180–207 (conditional cumulant closure), 273–306 (diagonal counterexample and lower certificate), 309–353 (tilted variance), 355–363 (dyadic scales), 365–400 (corner maps), and 402–442 (analytic errors and physical Gram matrices). The stronger limitation of its equation (16) is proved in Theorem 4.1; the original report was not edited.

The preceding working audit is C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/M151-proof-audit.txt. It records the public literature URLs and inspection boundaries reused here. The separately prepared local parameter audit was read in full: C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/M151-parameter-audit.txt. Its Sections 3–10 derive the fixed-power chain and Sections 11–12 register parameter limits and inherited inputs. Its follow-up Section 13 independently checks spectral continuity, the endpoint limit and a fixed bounded quadratic lower test. It found no new parameter-dependent obstruction; the geometric dependency in this report remains explicit. Failure of the direct $s = 2$ proof is not a failure of the separate conditional continuity argument in Corollary 9.8.

The separate local audit C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/M150-local-audit.txt was also read in full for its independently checked corner-moment calculation and exact involution. Its audit of M150 is not imported as validation of that manuscript here. Its stronger Parseval tail concerns an L^2 reciprocal-derivative norm, not the pointwise derivative enclosure of Theorem 7.1, so those tail estimates were not conflated.

For Proposition 6.3, the integer script and its data were read at the following exact paths:

C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/check_M151_Koebe_blocks.py
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_audit/check_M151_Koebe_blocks.json.

An independent in-memory replay recomputed all fourteen listed numerators, verified the strict fourth-power inequalities at depths 1–13 and the one-map upper test at 14. The input script was not executed or modified, and its JSON was not overwritten. This is an exact finite lower obstruction, not a numerical universal upper certificate.

The format/evidence comparator was C:/Users/mural/OneDrive/Documents/OpenAI-722-math/M149-Koebe-Circle-Domains/TR-M149-Ram-Koebe.tex. No figure-source or formal proof artifact was used as mathematical evidence. The private report's previously reported finite tests were not rerun. The proofs in this report establish the new implications; finite arithmetic or quadrature checks, when used during preparation, are only regressions, not substitutes for those proofs. No original source or delivery record was modified to create this report.

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